

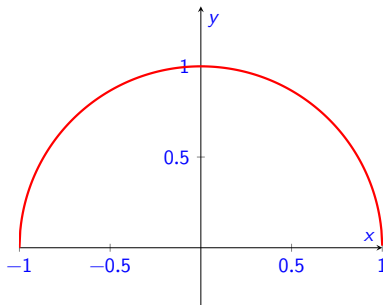
2627-MA140 Engineering Calculus

Week 04, Lecture 3
The Chain Rule and Inverse Functions

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Links in today's lecture on the Chain Rule

1 Recall: Differentiation “rules”

2 Chain Rule

3 Composites of 3 or more functions

4 Inverse functions

■ Inverse Rule

5 Implicit differentiation

6 Exercises

See also: Sections 3.6 (The Chain Rule) and 3.8 (Implicit Differentiation) of **Calculus** by Strang & Herman:

[https://math.libretexts.org/Bookshelves/Calculus/Calculus_\(OpenStax\)](https://math.libretexts.org/Bookshelves/Calculus/Calculus_(OpenStax))

Recall: Differentiation “rules”

Yesterday we learned:

1. Constant Function Rule If c is constant, then $(c)' = 0$.

2. Power Rule $(x^n)' = nx^{n-1}$.

3. Constant multiple rule $(cf(x))' = cf'(x)$

4a. Sum Rule $(f(x) + g(x))' = f'(x) + g'(x)$.

4a. Difference Rule $(f(x) - g(x))' = f'(x) - g'(x)$.

5. Product Rule $(u(x)v(x))' = u(x)v'(x) + v(x)u'(x)$.

6. The Quotient Rule $\left(\frac{u(x)}{v(x)}\right)' = \frac{v(x)u'(x) - u(x)v'(x)}{v(x)^2}$.

Chain Rule

Of all the differentiation rules, the **chain rule** is the most important: most other rules are actually just special cases of it. It applies to a “function of a function”

The Chain Rule

If $u(x)$ and $v(x)$ are differentiable, and f is the composite function $f(x) = u(v(x))$, then

$$\frac{df}{dx} = \frac{du}{dv} \frac{dv}{dx} \quad \Rightarrow \quad f' = u' v'$$

Note: $u(v(x))$ is not $u(x)v(x)$,
we read it as “ u of v of x ”.

Ex. If $f(x) = (2x+1)^3 = u(v(x))$ then

$$u(v) = v^3$$

$$v(x) = 2x+1.$$

$$\text{So } \frac{du}{dv}(v) = 3v^2$$

$$\frac{dv}{dx}(x) = 2. \quad \text{So } f'(x) = 3v^2(2) = 6(2x+1)^2$$

Chain Rule

The Chain Rule

If $f(x) = u(v(x))$, then

$$\frac{df}{dx} = \frac{du}{dv} \frac{dv}{dx}.$$

Example: What is the derivative of $f(x) = \cos(x^2)$?

$$u(v) = \cos(v)$$

$$v(x) = x^2$$

$$\frac{du}{dv} = -\sin(v)$$

$$\frac{dv}{dx} = 2x$$

$$\text{So } \frac{df}{dx} = \frac{du}{dv} \cdot \frac{dv}{dx} = -\sin(v) \cdot 2x = -2x(\sin(x^2))$$

Chain Rule

Example (To be read later)

Find $\frac{dy}{dx}$ if $y = (x^3 + 4x^4 + 7)^{12}$.

Example: Let $u(v) = v^{12}$ and $v(x) = x^3 + 4x^4 + 7$, then y is $y = u(v(x))$.

Note that

$$\frac{du}{dv} = 12v^{11} \quad \text{and} \quad \frac{dv}{dx} = 3x^2 + 16x^3.$$

By the Chain Rule we have

$$\frac{dy}{dx} = \frac{du}{dv} \frac{dv}{dx} = 12v^{11}(3x^2 + 16x^3),$$

and therefore

$$\frac{dy}{dx} = 12(x^3 + 4x^4 + 7)^{11}(3x^2 + 16x^3).$$

Chain Rule

Example (To be read later)

Find $\frac{dy}{dx}$ if $y = \frac{1}{(x^4 + 2x^2 + 8)^{40}}$.

We have $y = (x^4 + 2x^2 + 8)^{-40}$. We can write y as $y(x) = u(v(x))$ with

► $u(v) = v^{-40}$ and so $\frac{du}{dv} = -40v^{-41}$; and

► $v(x) = x^4 + 2x^2 + 8$, so $\frac{dv}{dx} = 4x^3 + 4x$.

Applying the Chain Rule: $\frac{dy}{dx} = \frac{du}{dv} \frac{dv}{dx}$, we get

$$\frac{dy}{dx} = -40v^{-41}(4x^3 + 4x) = \frac{-40(4x^3 + 4x)}{(x^4 + 2x^2 + 8)^{41}}$$

Composites of 3 or more functions

One can apply the **Chain Rule** to “functions of functions of functions”: if $y(x) = t(u(v(x)))$, then

$$\frac{dy}{dx} = \frac{dt}{du} \frac{du}{dv} \frac{dv}{dx}$$

Example

Find $\frac{dy}{dx}$ when $y = \sin^4(x^5 + 7)$. $= (\sin(x^5 + 7))^4$

$$t(u) = u^4$$

$$\frac{dt}{du} = 4u^3$$

$$u(v) = \sin(v)$$

$$\frac{du}{dv} = \cos(v)$$

$$v(x) = x^5 + 7$$

$$\frac{dv}{dx} = 5x^4$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{dt}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx} = 4u^3 \cdot \cos(v) \cdot 5x^4 \\ &= 4(\sin(v))^3 \cdot \cos(x^5 + 7) \cdot 5x^4 \\ &= 20x^4 \cdot \sin^3(x^5 + 7) \cos(x^5 + 7) \end{aligned}$$

Composites of 3 or more functions

Example

Show that the derivative of $y = \cos^2(1/x)$ is

$$\frac{dy}{dx} = 2 \frac{\sin(1/x) \cos(1/x)}{x^2}.$$

$$y = [\cos(1/x)]^2$$

$$t(u) = u^2$$

$$u(v) = \cos(v) \quad v(x) = x^{-1}$$

$$\begin{aligned} \frac{dt}{du} &= 2u = 2(\cos(v)) \\ &= 2\cos(1/x) \end{aligned}$$

$$\begin{aligned} \frac{du}{dv} &= -\sin(v) \\ &= -\sin(1/x) \end{aligned}$$

$$\frac{dv}{dx} = (-1)x^{-2}$$

$$\frac{dy}{dx} = \frac{dt}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx} = 2\cos(1/x) \cdot \sin(1/x) \cdot \frac{1}{x^2} \quad \checkmark$$

Inverse functions

Suppose that $y = f(x)$. That is, f maps x to y .

Then the **inverse** of f is the function, f^{-1} , that maps y back to x .

Example

▶ The inverse of $f(x) = \frac{1}{2}x$ is $f^{-1}(x) = 2x$.

▶ The inverse of $f(x) = \sqrt{x}$ is $f^{-1}(x) = x^2$.

Warning: $f^{-1}(x)$ is not the same as $\frac{1}{f(x)}$.

$$\rightarrow f(f^{-1}(x)) = f(2x) = \frac{1}{2}(2x) = x \quad \checkmark$$

$$f^{-1}(f(x)) = f^{-1}\left(\frac{1}{2}x\right) = 2\left(\frac{1}{2}x\right) = x$$

$$\rightarrow f(f^{-1}(x)) = f(x^2) = \sqrt{x^2} = x \quad \checkmark$$

It is often useful to be able to express the derivative (assuming there is one) of an inverse function $f^{-1}(x)$ in terms of the derivative of $f(x)$.

To do this, we use the following rule:

Inverse-Function Rule

If $y = f^{-1}(x)$, then $x = f(y)$ and also

$$\underbrace{(f^{-1})'(x)}_{\text{find}} = \frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{1}{f'(y)} = \frac{1}{f'(f^{-1}(x))}.$$

derivative of $f^{-1}(x)$

$$y = f^{-1}(x)$$

Example

If $y = x^{1/3}$, use the **Inverse Rule** to find $\frac{dy}{dx}$.

Note: we can solve this just using the **Power Rule**: $\frac{dy}{dx} = \frac{1}{3} x^{-2/3}$.

But we'll also do this with the **Inverse Rule** for purposes of *exposition*.

If $y = x^{1/3}$, then $y^3 = x$, or $x = y^3$ so

$$\frac{dx}{dy} = 3y^2. \quad (\text{Power Rule})$$

By the inverse rule,

$$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \left(\frac{1}{3y^2} \right)$$

As $y = x^{1/3}$ we have

$$\frac{dy}{dx} = \frac{1}{3(x^{1/3})^2} = \frac{1}{3} x^{-2/3}. \quad \checkmark$$

Example

Find the derivative of $\sin^{-1}(x)$

Let $y = \sin^{-1}(x)$, then $x = \sin(y)$ $(\star)$, so

$$\frac{dx}{dy} = \cos(y). \quad (\star\star)$$

From $\sin^2(y) + \cos^2(y) = 1$, we find $\cos(y) = \sqrt{1 - \sin^2(y)}$
(choosing the positive square root as $\cos(y)$ is positive for y here).
Using $(\star)$:

$$\cos y = \sqrt{1 - x^2}.$$

Now using the inverse rule and $(\star\star)$, we have

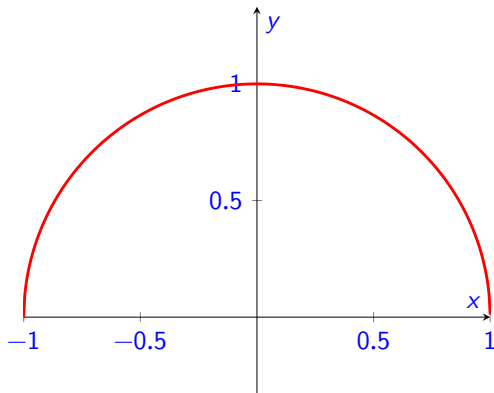
$$\frac{dy}{dx} = \frac{1}{\cos(y)} = \frac{1}{\sqrt{1 - x^2}}.$$

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Implicit differentiation

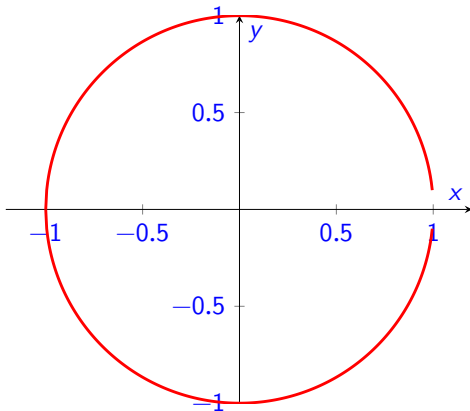
To date, most functions we have studied have been **explicitly** defined. Such functions can be written as $y = f(x)$: given a value of x we can substitute it into $f(x)$ to get the corresponding value of y .

Example: $y = \sqrt{1 - x^2}$.



Implicit differentiation

However, sometimes we are given an equation involving x and y where these two terms are not “separated” entirely; e.g., $x^2 + y^2 = 1$. Here y is **implicitly** defined: for any pair (x, y) we can check if it is on the curve described by the equation.



Implicit differentiation

Since **implicit equations** define curves, we can use **implicit differentiation**, for example, finding tangents to these curves.

Method:

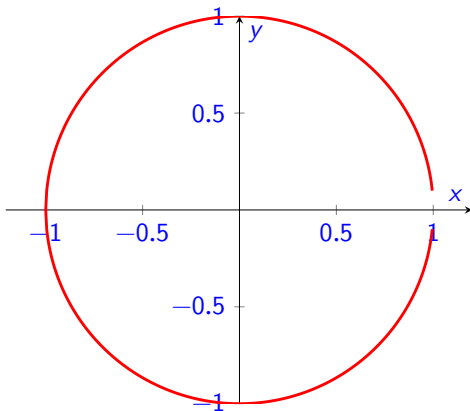
1. Differentiate both sides of the equation, with respect to x , keeping in mind that y is a function of x , using the Chain Rule where needed.
2. Solve for dy/dx .

Implicit differentiation

If y is defined by $x^2 + y^2 = 1$, find $\frac{dy}{dx}$.

Implicit differentiation

Now we know that if $x^2 + y^2 = 1$, then $\frac{dy}{dx} = -\frac{x}{y}$. We can check that this relates to the slope of the tangents to this curve at various places:



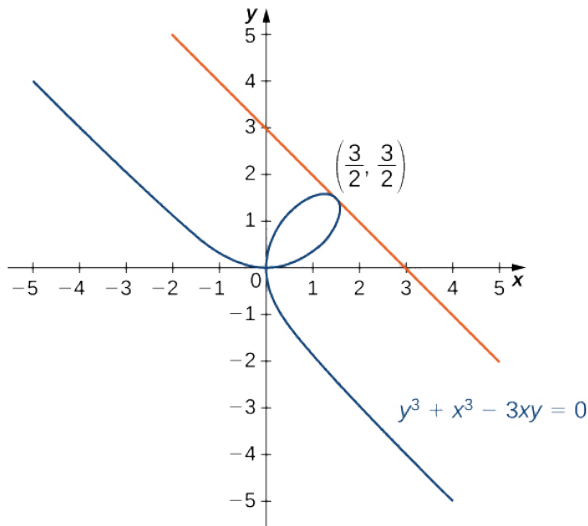
Implicit differentiation

Find the tangent to the curve $x^2 + y^2 = 25$, at the point $(3, -4)$.

Implicit differentiation

Find the tangent to the curve $y^3 + x^3 - 3xy = 0$, at the point $(3/2, 3/2)$.

Implicit differentiation



Exercises

Exercise 4.3.1

Find the derivative of

1. $f(x) = x^3 \cos(x^2)$
2. $f(x) = \tan^3(\sin^2(x^4))$

Exercise 4.3.2

Show that $\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$.

Exercise 4.3.3

Find the equation of the tangent to the curve defined by $x^2 - y^2 = 16$ at the point $(5, 3)$.